

First Order ODEs

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$$\left. \begin{array}{l} y' = f(x, y) \\ y(x_0) = y_0 \end{array} \right\} \boxed{\text{IVP}}$$

→ $\boxed{\text{Normal Form of an ODE}}$

$$F:$$

$$F(x, y)$$

$$dF =$$

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} \\ (x, y) \neq (0, 0) \end{cases}$$

$$f(0, 0) = 0$$

$$f_x(0, 0) = ? \quad f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(0+h, 0) - f(0, 0)}{h}$$

$$f_y(0, 0) = ?$$

$$= \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h}$$

$$f_x(a, b) = ?$$

$$= \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

$$(a, b) \neq (0, 0)$$

$$f(a+h, b+k) - f(a, b) = f_x(a, b) \cdot h + f_y(a, b) \cdot k = o(h, k)$$

$$y = mx$$

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$$

$$\{y = mx\}$$

$$= \text{DNE}$$

$$F: F(x, y)$$

Total Differential

$$dF = F_x dx + F_y dy$$

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy$$

Linear First Order ODEs

- An ODE that can be written in the form

$$a_0(x)y' + a_1(x)y = f(x),$$

is called a linear first order ODE. If $f(x) = 0$, the above ODE is called homogeneous. We also assume that a_0, a_1, F are continuous functions in an open interval.

- If we further assume that $a_0(x) \neq 0$ for all x in the interval then after dividing by $a_0(x)$ through out we get

$$y' + \frac{a_1(x)}{a_0(x)}y = \frac{f(x)}{a_0(x)}.$$

- We can also write the above equation in the form of

$$y' + b_1(x)y = F(x),$$

where $b_1(x) = \frac{a_1(x)}{a_0(x)}$ and $F(x) = \frac{f(x)}{a_0(x)}$.

Integrating Factor

- We now develop a technique that can be used to solve any first-order linear ODE that has been written in the form

$$\frac{d}{dx}(uy) = uy' + p(x)y = q(x)u \quad (1)$$

up to a couple of integrals. The key idea is to put linear ODEs in a form where we can integrate both sides with respect to x .

- If we differentiate uy where u and y are functions of x , we get

$$\frac{d}{dx}(uy) = uy' + u'y$$

$$u' = up(x)$$

- Multiplying the linear differential equation (1) by u gives us

$$uy' + up(x)y = uq(x). \quad [u(x)y'(x) + u(x)p(x)y'(x) = u(x)q(x)]$$

Integrating Factor

- The LHS of the last identity will be equal to $(uy)'$ if $\boxed{u' = up(x)}$.
- Then we can solve the separable ODE $\frac{du}{dx} = up(x)$. Solving it

$$\boxed{u = e^{\int p(x)dx}}$$

- With the above u our main equation takes the form

$$\boxed{\frac{d}{dx}(uy) = uq(x)}.$$

- Integrating this equation with respect to x , we obtain

$$uy = \int uq(x)dx + C.$$

- Solving for y , we deduce

$$\boxed{y = u^{-1} \int uq(x)dx + Cu^{-1}}.$$

$$= e^{-\int p(x)dx} \left(\int u q(x) dx \right) + C e^{-\int p(x)dx}$$

Example: Integrating Factor

Example

Solve the ODE

$$y' - \frac{1}{x}y = 0, \quad x > 0.$$

$$p(x) = -\frac{1}{x}$$
$$q(x) = 0$$

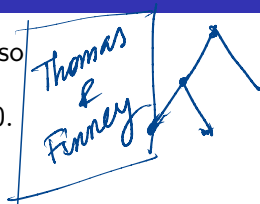
$$p(x) = -\frac{1}{x} \text{ and } q(x) = 0.$$

$$y = c e^{+\int \frac{1}{x} dx}$$

Exact ODE

- Another common form of a first order ODE is also

$$M(x, y)dx + N(x, y)dy = 0.$$



The above form is also called the differential form.

- These equations have a relation with implicit differentiation. If we use chain rule to differentiate $F(x, y) = c$ implicitly with respect to x , we get

$$F(x, y(x)) = c$$

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0. \quad (2)$$

- Thus if we have an equation of the form (2), we know that the solutions are given by the equation $F(x, y) = c$. In differential form (2) is written as

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = 0. \quad (3)$$

- Notice that the LHS of (3) is the total differential of F .

Exact ODEs

Definition (Exact Differential Equation)

The differential equation

$$M(x, y)dx + N(x, y)dy = 0$$

is said to be an exact in region R of the xy -plane if there exists a function $F(x, y)$ such that

$$\frac{\partial F}{\partial x} = M \text{ and } \frac{\partial F}{\partial y} = N, \quad (4)$$

for all (x, y) in R .

Example (An Exact ODE)

$$2x \sin y dx + x^2 \cos y dy = 0.$$

$M(x, y) = 2x \sin y$; $\frac{\partial M}{\partial y} = 2x \cos y$ Finding an 'F'
 $N(x, y) = x^2 \cos y$; $\frac{\partial N}{\partial x} = 2x \cos y$

Exact ODEs

Theorem

The general solution to an exact equation $M(x, y)dx + N(x, y)dy = 0$ is defined implicitly by $F(x, y) = c$ where F satisfies (4) and c is an arbitrary constant.

Theorem (Test for Exactness)

Let M, N and their first partial derivatives M_y and N_x be continuous in a region R of the xy -plane. Then the differential equation

$$M(x, y)dx + N(x, y)dy = 0$$

is exact for all x, y in R if and only if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Problem: Exact ODE

Example

Solve the following IVP:

$$(2x \cos y + 3x^2 y) dx + (x^3 - x^2 \sin y - y) dy = 0, \quad y(0) = 0.$$

$$M(x, y) dx + N(x, y) dy = 0, \quad \boxed{y(0) = 0}$$

$$M(x, y) = 2x \cos y + 3x^2 y$$

$$N(x, y) = x^3 - x^2 \sin y - y$$

$$\boxed{\frac{\partial F}{\partial x} = M}$$

$$(F(x, y) = ?)$$

$$\boxed{\frac{\partial F}{\partial y} = N}$$

$$\frac{\partial M}{\partial y} = -2x \sin y + 3x^2$$

$$\frac{\partial N}{\partial x} = 3x^2 - 2x \sin y$$

equation (ODE) is exact.

$$\frac{\partial F}{\partial x} = 2x \cos y + 3x^2 y$$

$$\Rightarrow F(x, y) = x^2 \cos y + x^3 y + \boxed{C}$$

$$= x^2 \cos y + x^3 y + g(y)$$

(where g is an arbitrary f^n in y)

Now $\frac{\partial F}{\partial y} = -x^2 \sin y + x^3 + \boxed{g'(y)}$

$$= x^3 - x^2 \sin y - y$$

$$\Rightarrow \boxed{g'(y) = -y}$$

~~⊗~~ $F(x, y) = x^2 \cos y + x^3 y - \frac{1}{2} y^2 + C$

$+ C = 0 \Rightarrow$ using IV. $\Rightarrow C = ?$

Non-Exact Equation

Definition (Integrating Factor)

A nonzero function $I(x, y)$ is called an integrating factor for the differential equation $M(x, y)dx + N(x, y)dy = 0$ if the differential equation

$$\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$$

is exact.

$$I(x, y)M(x, y)dx + I(x, y)N(x, y)dy = 0$$

$$M_1(x, y) = I(x, y)M(x, y) \quad N_1(x, y) = I(x, y)N(x, y)$$

Example

Show that $I(x, y) = x^2 y$ is an integrating factor for the differential equation

$$(3y^2 + 5x^2 y)dx + (3xy + 2x^3)dy = 0.$$

$$I(x, y) = x^m y^n$$

$m=2, n=1$

Remark

In some cases we can look for obtaining an integrating factor of the form

$$I(x, y) = x^m y^n \text{ where } m \text{ and } n \text{ are constants.}$$

$$\frac{\partial M}{\partial y} = 6y + 5x^2 \quad \frac{\partial N}{\partial x} = 3y + 6x^2$$

⊙ INTEGRATING FACTOR:-

Suppose $M dx + N dy = 0$ is not exact ; we want to find $I(x, y)$ so that

$(I(x, y)M(x, y)) dx + (I(x, y)N(x, y)) dy = 0,$
becomes exact.

Suppose we would like to find $I(x, y)$ depending only on x . Then,

$$\underline{I(x, y) = e^{\int g(x) dx}} \quad (\text{in terms of } x \text{ only})$$

$$\text{where } g(x) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}.$$

Similarly, if we want $I(x, y)$ to be in terms of y only, then

$$I(x, y) = e^{\int h(y) dy} \quad (\text{in terms of 'y' only})$$

where
$$h(y) = \frac{\partial M / \partial y - \partial N / \partial x}{-M}$$

$$M_0(x, y) = M(x, y) I(x, y)$$

$$N_0(x, y) = N(x, y) I(x, y)$$

$$\boxed{\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}} \quad (\text{non-exact})$$

$$\boxed{\frac{\partial M_0}{\partial y} = \frac{\partial N_0}{\partial x}} \quad \checkmark$$

Chapter 2, 9. Integrating
Factors
of Simmon's ODE book ✓

Homogeneous Function

Definition (Homogeneous Function)

A function $f(x, y)$ is called **homogeneous of degree n** if

$$\underline{f(tx, ty) = t^n f(x, y)}$$

for all t .

$f(x, y) = x + y$
+1
Non homogeneous
 $t=0$
 $f(0,0) = 0$

Example (Homogeneous Function)

- $x^2 + y^2$ is homogeneous of degree 2
- $x^2 + xy$ is homogeneous of degree 2
- $\frac{x+y}{x}$ is homogeneous of degree 0

$f(x, y) = \frac{xy}{x^2 + y^2}$
 $f(tx, ty) = \frac{(tx)(ty)}{(tx)^2 + (ty)^2}$

$t \neq 0$, $f(tx, ty) = t^0 f(x, y)$
← homogeneous of degree '0'

$(t \neq 0)$
 $= \frac{t^2}{t^2} \frac{xy}{x^2 + y^2}$
 $= xy / x^2 + y^2$

Homogeneous First Order ODE

Definition

A differential equation of the form

$$M(x, y)dx + N(x, y)dy = 0$$

is called an ODE with **homogeneous coefficients** if M and N are homogeneous functions of the same degree.

Remark

Such equations can be written as

$$\frac{dy}{dx} = H\left(\frac{y}{x}\right).$$

$$M dx + N dy = 0$$

— M, N are homogeneous of same degree.

$$\frac{dy}{dx} = - \frac{M(x, y)}{N(x, y)} = - \frac{M(x, x \cdot y/x)}{N(x, x \cdot y/x)} \quad (x \neq 0)$$

$$= - \frac{M(x \cdot 1, x \cdot y/x)}{N(x \cdot 1, x \cdot y/x)}$$

$$= - \frac{x^n M(1, y/x)}{x^n N(1, y/x)}$$

$$\Rightarrow \boxed{\frac{dy}{dx} = - \frac{M(1, y/x)}{N(1, y/x)}}$$

Let's say
 M and N both
are homogeneous
of degree n

Substitution Method

- To solve above ODE we use the substitution

$$v = \frac{y}{x}$$

- Then substitute

$$y = vx$$

- Differentiating, we get

$$\frac{dy}{dx} = v + x \frac{dv}{dx}.$$

- Substituting into the original equation reduces it to a **separable equation**.

Solution Procedure

Steps to solve ODE with homogeneous coefficients

① Verify that the equation is homogeneous.

② Introduce substitution

$$v = \frac{y}{x}$$

③ Write

$$y = vx, \quad \underline{\frac{dy}{dx} = v + x \frac{dv}{dx}}$$

④ Substitute into the equation.

⑤ Solve the resulting separable equation.

⑥ Replace $v = \frac{y}{x}$.

Example

Example (ODE with homogeneous coefficients)

Solve

$$\frac{dy}{dx} = \frac{x+y}{x}.$$

Rewrite

$$\frac{dy}{dx} = 1 + \frac{y}{x}.$$

Let

$$v = \frac{y}{x}, \quad y = vx.$$

Then

$$\frac{dy}{dx} = v + x \frac{dv}{dx}.$$

Example (Continued)

Substitute into the equation:

$$v + x \frac{dv}{dx} = 1 + v$$

$$x \frac{dv}{dx} = 1$$

$$\frac{dv}{dx} = \frac{1}{x}.$$

Integration

Separate variables

$$dv = \frac{dx}{x}$$

Integrate

$$v = \ln |x| + C.$$

Recall

$$v = \frac{y}{x}.$$

Hence

$$\frac{y}{x} = \ln |x| + C.$$

Therefore the final solution is

$$y = x \ln |x| + Cx.$$

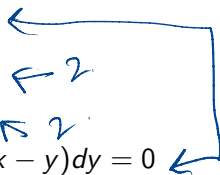
Practice Problems: ODE with Homogeneous Coefficients

Solve the following:

1 $\frac{dy}{dx} = \frac{x+y}{x-y}$

2 $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$

3 $(x+y)dx + (x-y)dy = 0$



Example:- Solve $(x \tan y/x + y) dx - x dy = 0$

$$M(x, y) = x \tan(y/x) + y; M(tx, ty) = (tx) \tan\left(\frac{ty}{tx}\right)$$

$$N(x, y) = x.$$

$\downarrow$
N is also homogeneous of order 1.

$$\frac{dy}{dx} = \frac{x \tan y/x + y}{x}$$

$$= \tan(y/x) + (y/x)$$

$$\boxed{u = y/x} \Rightarrow y = ux \Rightarrow x \frac{du}{dx} = \tan u + u$$
$$\Rightarrow x \frac{du}{dx} = \tan u \Rightarrow \int \frac{du}{\tan u} = \int \frac{dx}{x}$$

$$+ (ty) \\ = tx \tan(y/x) + ty \\ = t M(x, y)$$

$\downarrow$ $t \neq 0$
M is homogeneous of degree 1.

$$\boxed{y(x) = \dots}$$

First Order Nonlinear Equation: Bernoulli Equation

- We now consider a rather special type of equation that can be reduced to a linear equation by an appropriate transformation. This is the so-called Bernoulli equation.

Definition (Bernoulli Equation)

An equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad (5)$$

Handwritten note: $\frac{dy}{dx} + y = y^3$

is called a Bernoulli Equation.

Theorem

Suppose $n \neq 0, 1$. Then, the transformation $v = y^{1-n}$ reduces the Bernoulli equation (5) to a linear equation in v .

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

$$v = y^{1-n}$$

$$\frac{dv}{dx} = (1-n) \underbrace{y^{-n}} \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{1-n} y^n \frac{dy}{dx}$$

linear equation in 'v'
— i.f

Example: Bernoulli Equation

Example (Bernoulli Equation)

Solve $\frac{dy}{dx} + y = xy^3$.

$$n=3 \quad ; \quad v = y^{1-3} = y^{-2}$$

$$\Rightarrow \frac{dy}{dx} + y = xy^3 \quad \Rightarrow \quad \boxed{\frac{dv}{dx} + p(x)v = q(x)}$$

$$v = y^{-2}$$

$$\frac{dv}{dx} = - \dots$$

$$p(x) = -2$$

$$q(x) = -2x$$

$$\frac{dv}{dx} - 2v = -2x$$

$v \rightarrow$ using int. factor

$\rightarrow$ solve for y ;